Bioinformatics

BY

DR. ANUPAM GHOSH

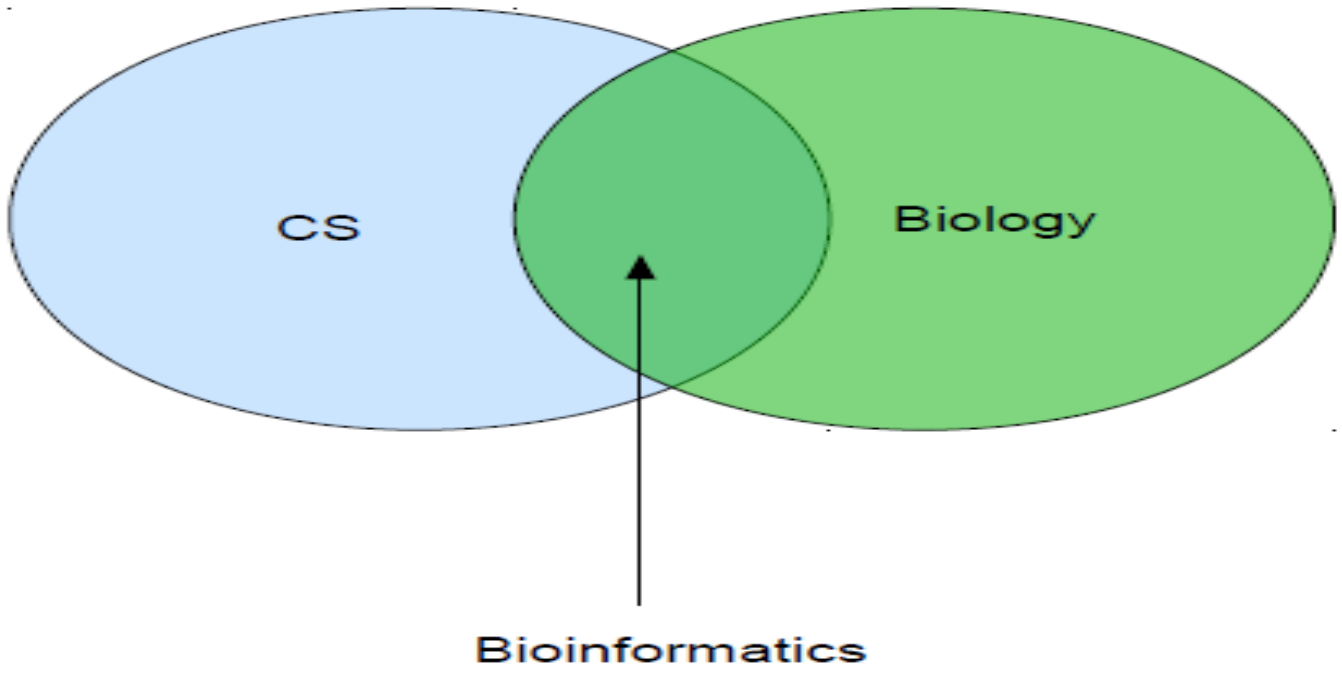
Email: anupam.ghosh@rediffmail.com

Professional Profile:
https://www.nsec.ac.in/fps/faculty.php?id=138

LinkedIn Profile:
https://www.linkedin.com/in/anupam-ghosh-1504273b/

Research Profile:
https://www.researchgate.net/profile/Anupam_Ghosh14

What is Bioinformatics?



Important problems → medical applications,
Infectious diseases, genetic defects etc.
Masses of data → storage and analysis challenges
HPC → increased need for parallel codes

What is bioinformatics?

- Bioinformatics, *n.* The science of **information and information flow in biological systems**, esp. of the use of computational methods in **genetics and genomics**. (Oxford English Dictionary)

- "The **mathematical**, **statistical** and **computing** methods that aim to solve biological problems using **DNA and amino acid sequences** and related information." -- Fredj Tekaia

What is Biological Raw Data?

- There are many types of biological raw data
  - Images from microscopes
  - Microarray data
  - Protein structure data
  - Morphological data
  - Ecological data
  - Biogeographical data
  - …
- In this course we will mainly focus on *classic* Bioinformatics, that is, the analysis of molecular sequence data (DNA, protein data)

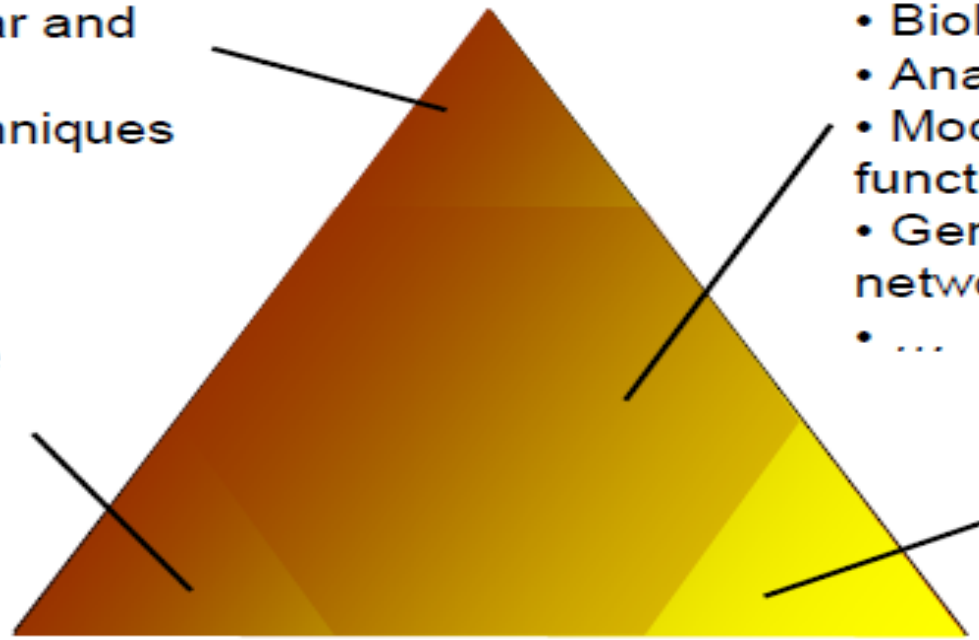
Computational biology

- Application of **computing** to **biology** (broad definition)
- Often used interchangeably with bioinformatics
- Or: ***Biology*** that is done with **computational means**

Bioinformatician's skill set

Biology & Medicine
- Basics in molecular and cell biology
- Measurement techniques

Bioinformatics
- Biological sequence analysis
- Biological databases
- Analysis of gene expression
- Modeling protein structure and function
- Gene, protein and metabolic networks
- …

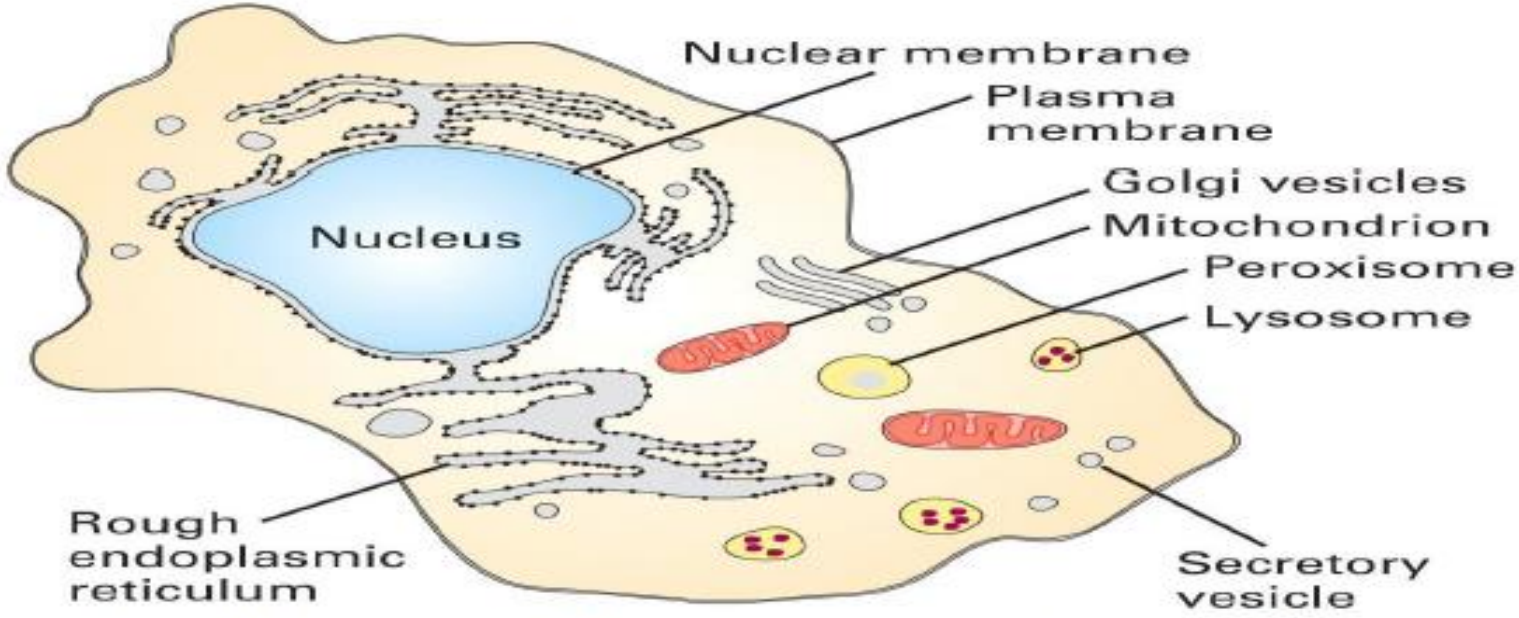


Computer Science
- Programming
- Databases
- Algorithmics

Mathematics and statistics
- Calculus
- Probability calculus
- Linear algebra

Where would you be in this triangle?

Life begins with Cell

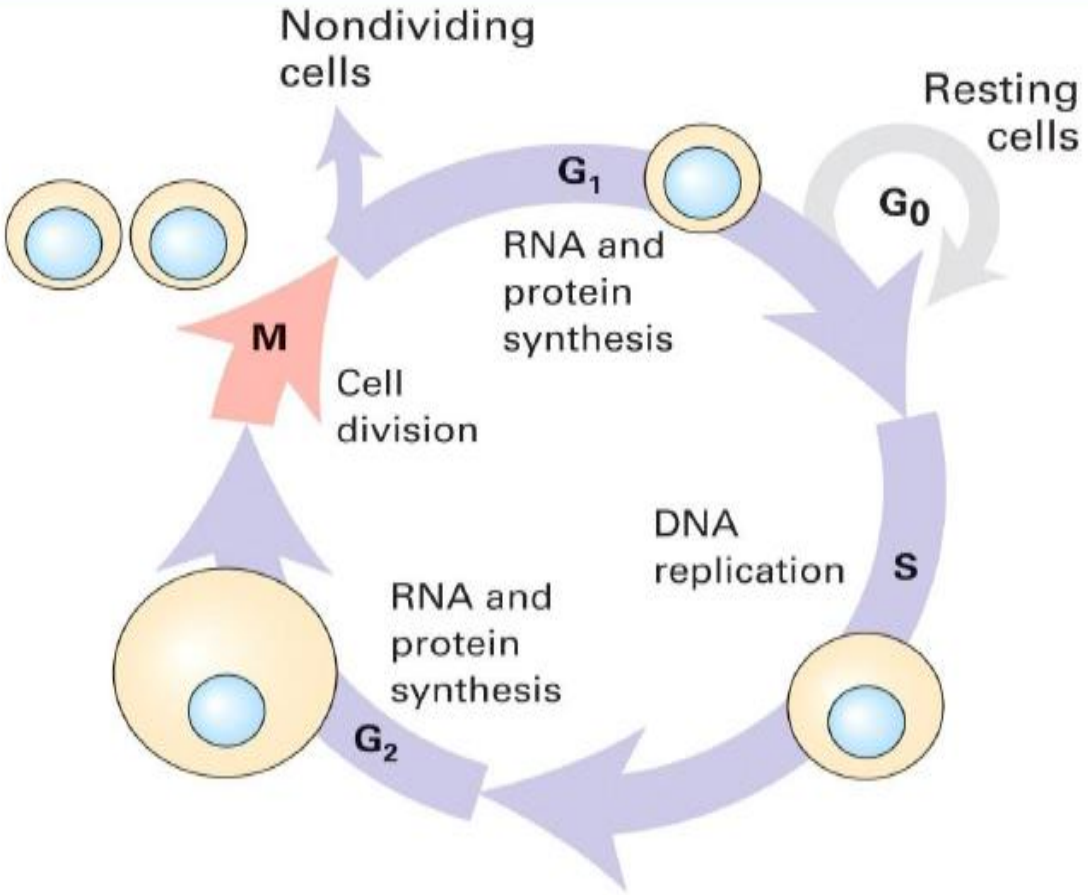


- A cell is a smallest structural unit of an organism that is capable of independent functioning
- All cells have some common features

Cells

- Fundamental working units of every living system.
- Every organism is composed of one of two radically different types of cells:
  - prokaryotic cells or
  - eukaryotic cells.
- Prokaryotes and Eukaryotes are descended from the same primitive cell.
  - All prokaryotic and eukaryotic cells are the result of a total of 3.5 billion years of evolution.

All Cells have common Cycles



- Born, eat, replicate, and die

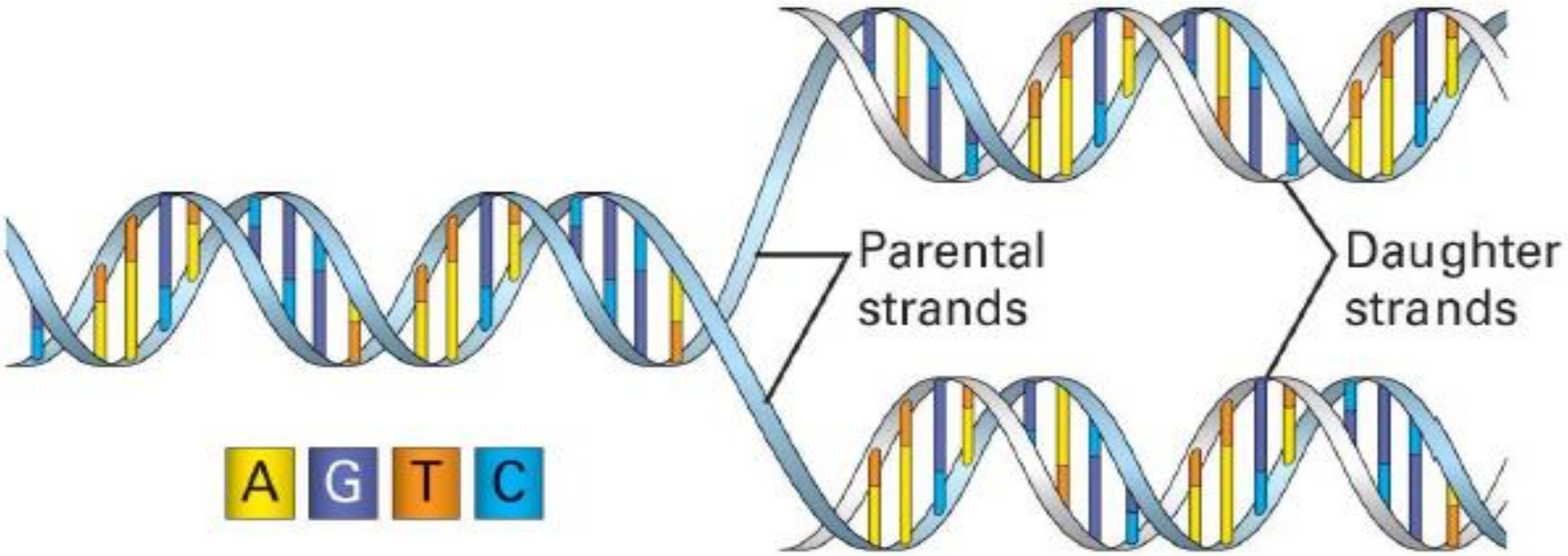
Common features of organisms

- Chemical energy is stored in ATP
- Genetic information is encoded by DNA
- Information is transcribed into RNA
- There is a **common triplet genetic code**
- Translation into proteins involves ribosomes
- Shared metabolic pathways
- Similar proteins among diverse groups of organisms

All Life depends on 3 critical molecules

- DNAs (Deoxyribonucleic acid)
  - Hold information on how cell works
- RNAs (Ribonucleic acid)
  - Act to transfer short pieces of information to different parts of cell
  - Provide templates to synthesize into protein
- Proteins
  - Form enzymes that send signals to other cells and regulate gene activity
  - Form body's major components (e.g. hair, skin, etc.)
  - "Workhorses" of the cell

DNA: The Code of Life

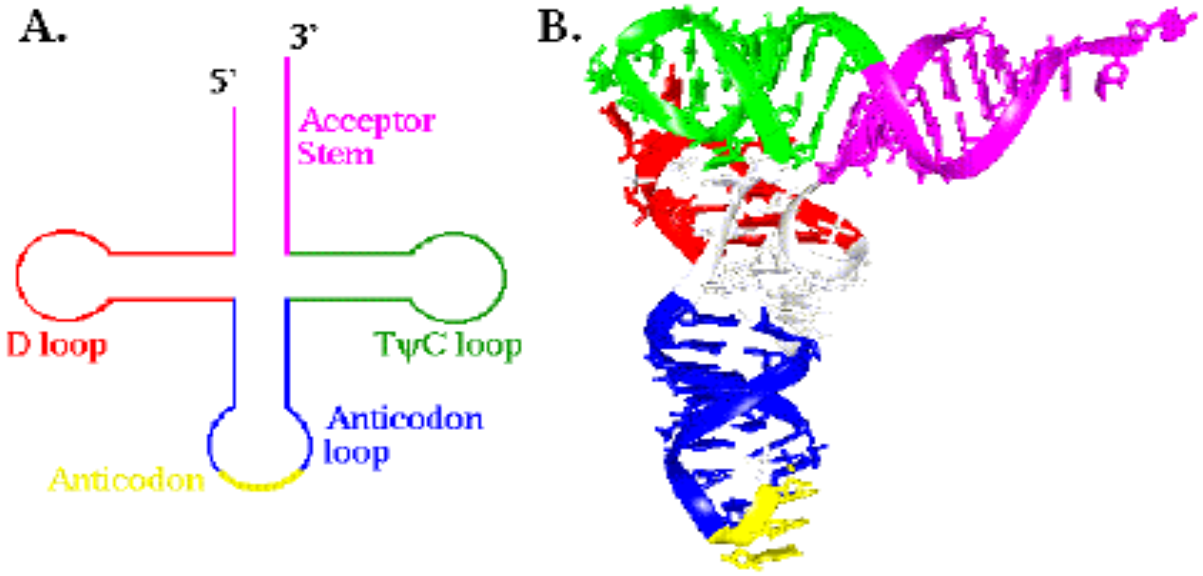


- The structure and the four genomic letters code for all living organisms
- Adenine, Guanine, Thymine, and Cytosine which pair A-T and C-G on complimentary strands.

► Lecture: Genome sequencing and assembly

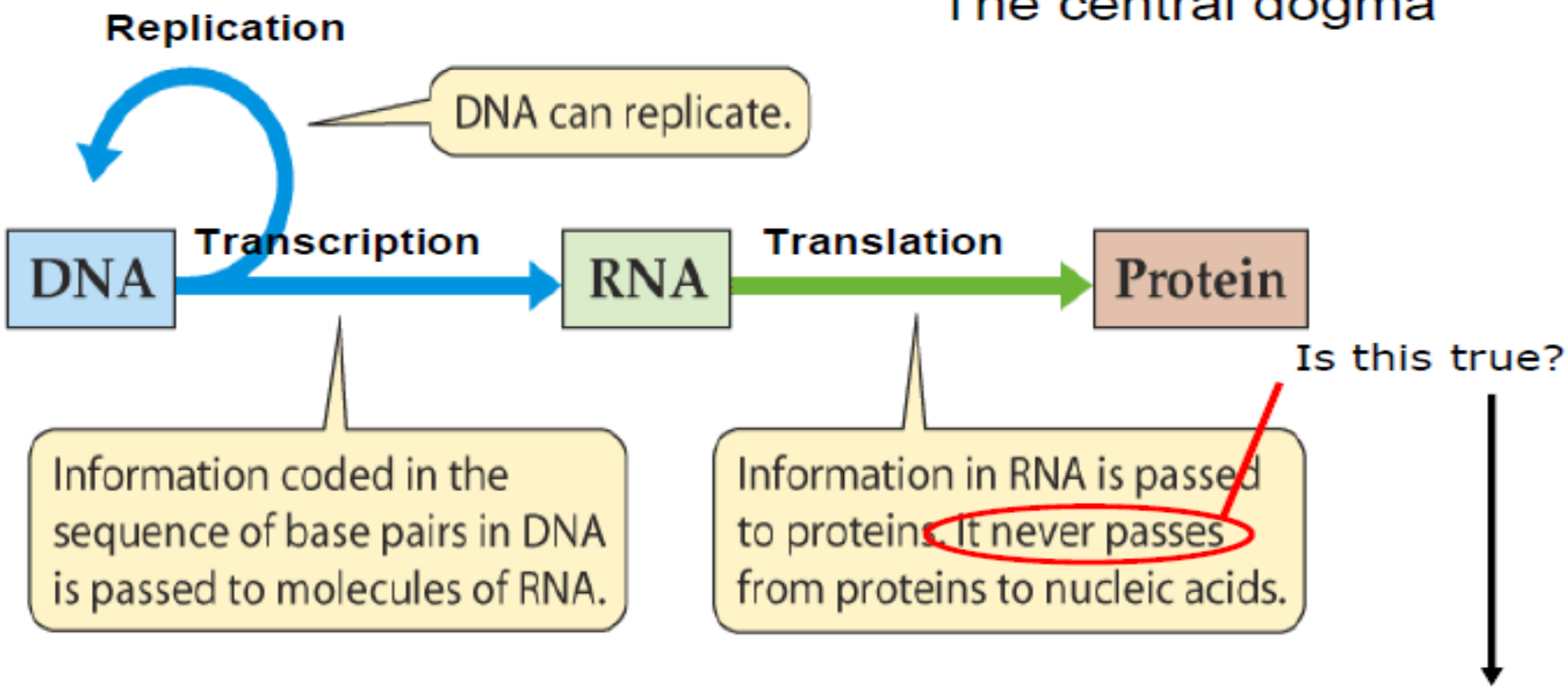
RNA

- RNA is similar to DNA chemically. It is usually only a single strand. T(hyamine) is replaced by U(racil)
- Several types of RNA exist for different functions in the cell.



DNA, RNA, and the Flow of Information

"The central dogma"



It will be Continued….